

Syed Taha

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EDUCATION

Institute of Business Administration (IBA)

Bachelor of Science in Computer Science, GPA: 3.53/4.0

Karachi, Pakistan

Expected May 2027

EXPERIENCE

Teaching Assistant

Institute of Business Administration, Karachi

Aug 2024 - Present

Karachi, Pakistan

- Supported Introduction to Programming & OOP, Computer Architecture and Operating Systems by facilitating weekly C++ labs for 100+ students and giving one-on-one guidance on problem-solving and debugging.
- Supervised student projects, from 10 Computer Architecture teams building RISC-V assembly and hardware-optimized systems (S4 models for galaxy morphology classification) to Operating Systems projects extending the xv6 LLM runtime into distributed inference, covering OS security, distributed inference security, and networked operating systems.
- Graded assignments and assessments and acted as liaison between faculty and students, clarifying requirements and keeping academic standards consistent.

Founding Member and Researcher

Systems Research @ IBA

Jul 2026 - Present

Karachi, Pakistan

- Conduct research on systems for ML inference with the group, including PiSSM (distributed state space model inference on a Raspberry Pi 5 cluster) and an AlphaZero-style Othello engine (MCTS guided by a self-play-trained CNN, distilled for deployment).
- Created the group's website and GitHub organization.

Data Science Intern

Foretheta, LLC

Aug 2026 - Sep 2026

Seattle, WA (Remote)

- Diagnosed and resolved data-deduplication bugs in a search reranking pipeline, improving result quality while preserving system guarantees around output completeness; added automated test coverage for the fix.
- Built and maintained a statistical evaluation framework for comparing search-ranking algorithm variants, using paired bootstrap confidence intervals to distinguish genuine improvements from statistical noise.
- Performed root-cause analysis of search ranking quality issues across a large evaluation dataset, tracing problems back to specific bugs in request construction and model input handling.
- Designed and ran a statistically rigorous comparison of alternative machine learning models for a domain-specific retrieval task, including a power analysis to size the evaluation appropriately and a licensing review of candidate models.

Software Engineering Fellow

Headstarter AI

Jul 2024 - Sep 2024

Long Island City, NY (Remote)

- Delivered 5 AI-powered web apps in 5 weeks using React/Next.js, Firebase, Clerk, and Vercel under weekly agile sprints with CI/CD, scaled each to 200+ users and launched an IBA teacher-review portal with 300 waitlist sign-ups.
- Co-developed a Next.js customer-support agent with a custom OpenAI+Pinecone RAG pipeline and a GPT-4o-powered flashcard SaaS, with weekly innovation sessions with engineers from Google, Y Combinator, Stanford, and Amazon.

PROJECTS

Project GLIDE: GPS-Denied Drone Navigation | *Python, PyTorch*

- Researching GPS-free drone navigation by magnetic field: whether a drone carrying only a 3-axis magnetometer can navigate a region using the Earth's magnetic field, supported by our own ground surveys.
- Investigating a small S4D state space model as the pilot in a physics-informed simulator, in closed loop with its own predicted position and heading as the next state, so that physical laws govern navigation instead of a purely machine learning approach.

LLM Inference in the xv6 Educational OS | *C, RISC-V, xv6*

- Engineered a POSIX-compliant Shared Memory (SHM) subsystem for the xv6 kernel to enable zero-copy, persistent caching of Large Language Model weights (about 60MB), eliminating redundant network transfers and reducing subsequent inference latency from about 60s to sub-second.

- Designed a cycle-accurate performance profiling library that tracks call hierarchies and identified matrix multiplication as the dominant bottleneck (consuming 87-92% of inference time), providing a data-driven basis for optimization.
- Successfully ported the LLaMA-2 inference engine to xv6, resolving system-level challenges (stack overflow via iterative quicksort) and replacing Linux dependencies, achieving stable throughput of about 16 tokens/sec on RISC-V.

Vectorized CNN Inference on 32-bit RISC-V Cores | *RISC-V Assembly*

- Implemented a complete Convolutional Neural Network in RISC-V assembly, leveraging the Vector Extension (RVV) to optimize core operations (convolution, pooling, softmax) for edge computing constraints.
- Achieved a 7.3x speedup in total inference time (8.223s to 1.122s) and a 12.2x reduction in convolution instructions via algorithmic redesign (im2col + GEMM) and FP16 compression.

PiSSM: Distributed State Space Model Inference | *Python, PyTorch, gRPC, FastAPI*

- Built a distributed inference system on six Raspberry Pi 5 boards (24 GB total, \$1,530) that runs Mamba-based large language models no single 4 GB board can hold.
- Split the model across boards with pipeline parallelism, at only 1.1 to 1.2x the latency of a single node.

Pariah: GNN-Transformer Intrusion Detection | *Python, PyTorch*

- Co-authoring a systematization-of-knowledge study of whether GNN-Transformer hybrids stay reliable for intrusion detection when the deployment network differs from the training network, with a taxonomy of graph-based detectors.
- Testing invariance objectives (IRM, EERM, DIR) against tuned ERM and Group DRO under pre-stated hypotheses, using 10+ seeds per configuration, paired bootstrap confidence intervals, and Holm correction.

Othello AI with MCTS & Optimized CNNs | *Python, PyTorch*

- Researched and developed a deep reinforcement learning AI by integrating Monte Carlo Tree Search (MCTS) with optimized Convolutional Neural Networks for board evaluation and move prediction.
- Systematically optimized the CNN architecture, reducing model size by 50% (85MB to 42.66MB) while increasing gameplay speed 10x (1000ms to 100ms per move) and maintaining a win rate over 90% against larger baselines.

OctaC: Statically Typed Language and Compiler | *C++, Python*

- Designing and implementing a compiler for a statically typed, C-structured language for numerical and matrix computation with built-in vector and matrix types.
- Completed the DFA-driven lexer and the parser with a pytest suite; semantic analysis and code generation follow, documented in a LaTeX technical book.

TECHNICAL SKILLS

Languages: C++, C, Python, RISC-V Assembly, Bash, JavaScript

Frameworks & Tools: PyTorch, React, Next.js, FastAPI, gRPC, Google Cloud, Git, GitHub Actions

High-Performance Computing: CUDA (Basic), Vectorization, Multi-threading, Performance Profiling

ACHIEVEMENTS

- Achieved 1st place in the “*Enhanced Safe Driver Prediction*” Kaggle competition among 100+ participants.
- Achieved 1st place in the “*Hello World*” competitive programming competition organized by CSS at IBA.